Mises's Fatal Flaw: Binding Deficits and the Memetic Evolution of Political Liberalism

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ABSTRACT

In *Human Action* (1949), Ludwig von Mises diagnosed a "fateful error" in classical liberalism: faith that rationality suffices for political persuasion. This paper translates Mises's diagnosis into memetic theory, revealing a dual structural deficiency: a *binding deficit* (psychological incapacity to generate identity adhesion independent of instrumental outcomes) and vulnerability to *entrenched power networks* (material interests with high binding and concrete resources). Liberalism optimized for propositional coherence while neglecting the psychological infrastructure for durable mass replication and the resistance of networks combining tribal loyalty with tangible benefits.

I distinguish between *memetic content* (propositions, institutions) and *transmission vehicles* (identity structures, rituals). Classical liberalism conflated these dimensions. Mises recognized the error but offered no solution without abandoning rationalism itself. Contemporary Latin American liberal movements have discovered a dual workaround: (1) hybridization with high-binding sources (evangelical narratives, religious symbolism); (2) explicit attacks on embedded networks. This separation of content from vehicle, combined with delegitimizing material power structures, represents a testable evolutionary innovation.

Six historical cases (French Revolution, Comte's positivism, Soviet Union, Kemalism, Argentine Enlightenment, contemporary Latin America) spanning two centuries and three continents reveal a universal pattern: rationalist projects, regardless of ideological content, face identical binding deficits and power network resistance. Evidence from Argentina and Brazil (2018-2025) shows liberal governments adopting religious outreach strategies coinciding with structural attacks on established political networks. Preliminary data suggest resilience to economic crises and cross-spectrum competition for religious binding, with actors from opposed ideological positions seeking evangelical constituencies.

The framework contributes to Extended Phenotype Theory by formalizing binding coefficients as institutional fitness predictors, explaining persistent dysfunction, and proposing testable predictions about hybridization dynamics.

Keywords: memetics, political liberalism, Ludwig von Mises, extended phenotype theory, binding mechanisms, religious-political hybridization, Latin American politics, institutional evolution

JEL Codes: D72 (Political Processes), P16 (Political Economy), Z12 (Religion), Z13 (Social Norms), B53 (Austrian Economics)

I. THE PARADOX

1.1 Two admissions of failure

Bernardino Rivadavia resigned the presidency of Argentina on June 27, 1827. His modernization program, meticulously designed according to Enlightenment principles, had collapsed. Provincial caudillos rejected his Constitution. The Catholic Church resisted his ecclesiastical reforms. The legislature voted down his most ambitious projects. In his resignation speech, Rivadavia confessed: "Fatal es la ilusión en que cae un legislador cuando pretende que sus talentos y voluntades pueden mudar la naturaleza de las cosas." (*Fatal is the illusion into which a legislator falls when he pretends that his talents and will can change the nature of things*.)

Rivadavia's confession captures both dimensions of the problem: "talentos y voluntades" (rational design, the binding deficit) could not override "la naturaleza de las cosas" (embedded power structures with concrete interests). The caudillos possessed not only tribal loyalty from their gaucho militias (high binding) but also territorial control, tax collection, and military force (material networks). The Church held not only the faithful's devotion (binding) but also land, educational monopoly, and social legitimacy (institutional power). Rational arguments could overcome neither dimension.

One hundred twenty-two years later, Ludwig von Mises published *Human Action* (1949). In Chapter IX, analyzing the role of ideas in history, Mises identified what he called the "fateful error" of liberal thought:

> "This prejudice was the fateful error of the humanitarians, the philosophers, and the liberals. Men are not infallible; they err very often. It is not true that the masses are always right and know the means for attaining the ends aimed at. 'Belief in the common man' is no better founded than was belief in the supernatural gifts of kings, priests, and noblemen." (Mises, 1949, p. 193)

The error was not epistemological but anthropological. Liberals had replaced one unfounded myth (divine right of kings) with another (wisdom of crowds). Democracy, Mises warned, "cannot prevent majorities from falling victim to erroneous ideas and from adopting inappropriate policies which not only fail to realize the ends aimed at but result in disaster. Majorities too may err and destroy our civilization." (Mises, 1949, p. 193)

Two men, separated by a century, continent, and language, reached the same conclusion: rationalism alone cannot sustain a political program. Rivadavia experienced it empirically; Mises formulated it theoretically. Neither could resolve it without abandoning the rationalist identity that defined their projects.

1.2 The puzzle

Why did liberalism ignore Mises's warning for 60 years (1949-2008)? Why did contemporary Latin American liberal leaders, a century after Rivadavia, discover a solution that earlier generations could not?

The answer lies in evolutionary dynamics. Between 1949 and 2008, memetic selection pressure on liberalism was suspended. The Cold War allowed liberals to define themselves negatively (anti-communism) without generating positive binding. The Washington Consensus (1989-2008) eliminated serious ideological competition. Liberalism could survive in elite niches that did not require mass mobilization.

The 2008 financial crisis and the rise of populisms restored selection pressure. Liberalism again faced competition from programs with high binding: chavismo, trumpismo, various nationalisms. Party identification declined across Latin America between 2006 and 2018, with classical liberal parties rarely achieving significant parliamentary representation until recent electoral breakthroughs.

1.3 The thesis

Rational political programs suffer from what I call a *binding deficit*: structural incapacity to generate identity adhesion independent of instrumental outcomes. This deficit is not accidental but constitutive. Rationalism defines liberalism's identity, not merely its tactics.

The solution contemporary leaders appear to have found, hybridization with religious narratives, represents an evolutionary innovation that separates programmatic content from transmission vehicle. The content travels in a container not its own.

1.4 Contribution

This paper tries to contribute to Extended Phenotype Theory (EPT) in three dimensions:

Conceptual: Distinguishes memetic content from transmission vehicles as analytically separable variables.

Metric: Proposes "binding coefficient" as a predictor of institutional fitness.

Empirical: Offers testable predictions about the evolution of political liberalism in conditions of restored memetic competition.

II. THEORETICAL FRAMEWORK

2.1 The binding deficit formalized

Jonathan Haidt's *The Righteous Mind* (2012) distinguishes between "individualizing" moral foundations (care, fairness) and "binding" foundations (loyalty, authority, sanctity).

Liberalism appeals almost exclusively to the former. This is not tactical error but identity definition.

Definition: The *binding coefficient* β of a political program is the proportion of its appeal deriving from identity adhesion versus instrumental expectations.

$$\beta = \frac{\text{Identity adhesion}}{\text{Identity adhesion + Instrumental expectations}}$$

- High binding (β → 1): "I am Peronist because my family always was Peronist."
- Low binding (β → 0): "I vote liberal because I believe it will reduce inflation."

High-binding adhesion survives instrumental failures. Low-binding adhesion does not.

2.2 Memetic content versus transmission vehicles

Classical liberalism conflated two distinct properties:

Property	Definition	Liberal example
Truth	Correspondence with reality	"Markets allocate resources efficiently"
Fitness	Replication capacity	Low: requires sophisticated economic understanding

Liberals assumed truth implied fitness. But memetic selection operates on replication capacity, not correspondence with reality. A false but sticky meme outcompetes a true but complex one.

The separation principle: Content and vehicle are analytically distinct. Content can travel in vehicles not originally its own.

- **Content:** Propositions, rules, institutions (e.g., free markets, property rights, rule of law)
- **Vehicle:** Identity structures, rituals, belonging narratives (e.g., evangelical community, nationalist narrative, tribal loyalty)

Rivadavia's project had content but no vehicle. The Generation of 1837 (Echeverría, Alberdi, Sarmiento) attempted to supply one through Romantic nationalism. Sarmiento's *Facundo* (1845) is a liberal program wrapped in epic narrative: civilization versus barbarism. The content remained rationalist; the vehicle became emotional.

2.3 Integration with Extended Phenotype Theory

In the EPT framework, institutions function as extended phenotypes of the memes that generate them. A high-binding meme produces more resilient institutions because its support base does not depend on institutional performance.

This explains the central paradox of my prior work: why dysfunctional institutions persist while "rational" reforms fail. The former have binding; the latter do not.

Formal statement: Institutional fitness F is a function of both functional performance P and binding coefficient β:

$$F(\text{institution}) = f(P, \beta)$$

where $\frac{\partial F}{\partial \beta} > 0$ even when $\frac{\partial F}{\partial P} < 0$.

An institution can have negative performance (P < 0) yet positive fitness (F > 0) if β is sufficiently high. This is why clientelism outcompetes meritocracy in contexts where tribal loyalty (high β) dominates rational evaluation (low β).

2.4 Why Mises could not solve the problem

Mises's recognition of the binding deficit emerged gradually and painfully. In *Liberalism* (1927), written before the Nazi catastrophe, he defended rationalism against its critics:

> "Liberalism is usually reproached, besides, for being rationalistic. It wants to regulate everything reasonably and thus fails to recognize that in human affairs great latitude is, and, indeed, must be, given to feelings and to the irrational generally: i.e., to what is unreasonable." (Mises, 1927, p. 5)

His defense was normative rather than descriptive: "Liberalism does not say that men always act intelligently, but rather that they *ought*, in their own rightly understood interest, always to act intelligently." (Mises, 1927, p. 5, emphasis added)

Between 1927 and 1949, Mises witnessed the collapse of precisely the societies he had defended. Hitler's rise (1933), the Anschluss (1938), the confiscation of his Vienna library by National Socialists, his flight to Geneva (1934) and then New York (1940). The Weimar Republic's educated middle class chose Hitler. Austria's cosmopolitan capital welcomed annexation.

By 1949, Mises no longer defended rationalism as an unfulfilled ideal. He diagnosed it as a "fateful error": not because reason failed as a method, but because liberals overestimated humanity's commitment to using it. Yet he offered no solution because solving the problem required abandoning the rationalist identity that defined his entire intellectual project.

Mises could not advocate for non-rational transmission mechanisms without contradicting his life's work. His conclusion about liberalism's future was conditional but stark: "It may be that this optimism is unfounded and that the liberals have erred. But then there is no hope left for mankind's future." (Mises, 1949, p. 157)

This was not rhetorical despair but logical consistency. *If* humans are not primarily rational *and* liberalism depends on rationality, *then* liberalism has no viable path forward. Mises did not anticipate content-vehicle separation because it violated the unity of rational discourse he defended. For Mises, the *how* of transmission (rational persuasion) was inseparable from the *what* (liberal content). Hybridization would have seemed like contamination.

III. HISTORICAL PRECEDENT: THE ARGENTINE SEQUENCE

3.1 Rivadavia's failure (1821-1827)

Bernardino Rivadavia served as chief minister under Governor Martín Rodríguez (1821-1824) and then as Argentina's first president (1826-1827). His program included:

- Ecclesiastical reform (reducing Church power, suppressing monasteries)
- Public education expansion (founding Universidad de Buenos Aires, 1821)
- Public credit (negotiating Baring Brothers loan, 1824)
- Electoral reform (expanding suffrage)
- Institutional modernization (European-model bureaucracy)

Every component was rationally designed. None had binding mechanisms. The provinces rejected his centralizing Constitution of 1826. Caudillos like Juan Facundo Quiroga and Estanislao López mobilized gaucho militias. The unitario-federal civil war erupted. Rivadavia resigned on June 27, 1827, after 17 months in office.

His confession captures the binding deficit: rational design ("talentos y voluntades") cannot override embedded social structures ("la naturaleza de las cosas").

3.2 The Romantic correction (1837-1880)

The Generation of 1837 (Generación del 37) understood Rivadavia's error. Oscar Terán, in *Historia de las ideas en Argentina* (2008), describes their response:

> "Allí donde [la Ilustración] colocaba como núcleo de intelección de la realidad a la razón según el modelo de la ciencia físico-matemática, el romanticismo atiende a fenómenos que no forman parte del ámbito de la racionalidad iluminista. Así cobrará relieve la exploración de los aspectos irracionales de la conducta humana, tales como la imaginación o el ámbito de las emociones." (Terán, 2008, p. 54)

Translation: Where Enlightenment placed reason modeled on physical-mathematical science as the core of reality's intelligibility, Romanticism attends to phenomena outside rationalist scope, exploring irrational aspects of human conduct such as imagination and emotions.

Esteban Echeverría's *Dogma Socialista* (1846), Juan Bautista Alberdi's *Bases* (1852), and Domingo Faustino Sarmiento's *Facundo* (1845) retained Rivadavia's modernizing content but wrapped it in emotional vehicles:

- National epic (civilization versus barbarism)
- Local color (gaucho culture, pampas landscape)
- Existential struggle (the Republic's survival depends on defeating caudillismo)

Sarmiento's *Facundo* is the paradigmatic case. Subtitle: *Civilización y Barbarie* (Civilization and Barbarism). The program is Enlightenment (education, immigration, railroads, European institutions). The vehicle is Romantic (epic biography of the caudillo Facundo Quiroga, vivid descriptions of gaucho life, moral drama framed as cosmic struggle).

The content is liberal. The vehicle is emotional. The hybridization worked: Sarmiento became president (1868-1874) and implemented much of Rivadavia's program.

3.3 The recurrence of the pattern

Period	Rational project	Binding deficit	Compensation attempt
1821-1827	Rivadavia	Total	None (failure)
1837-1880	Generation of 1837	Recognized	Romantic nationalism
2023-present	Austrian liberalism	Recognized	Religious hybridization

Argentina has faced this problem three times in 200 years. The difference is that today the available binding source is not Romantic nationalism (occupied by Peronism) but expanding evangelicalism.

IV. A UNIVERSAL PATTERN: SIX CASES ACROSS TWO CENTURIES

4.1 The universality hypothesis

If the binding deficit were specific to liberalism or Latin America, we would have an interesting but limited phenomenon. Historical evidence suggests something deeper: every rationalist project, regardless of programmatic content, faces the same structural problem. Reason alone does not generate durable adhesion.

I present six cases distributed across two centuries, three continents, and opposed political programs. What they share is the sequence: (1) pure rational project; (2) failure or insufficient adhesion; (3) incorporation of high-binding elements, typically religious or pseudo-religious.

4.2 Case 1: The French Revolution (1789-1794)

Phase 1 - Cult of Reason (1793): The Hébertists, radical faction, established an explicitly atheist cult. They transformed Notre-Dame into a "Temple of Reason" on November 10, 1793. The declared objective: celebrate liberty, rationalism, empirical truth. An anthropocentric religion rejecting all deity.

Failure: Even other revolutionaries considered it excessive. Maximilien Robespierre denounced it as "ridiculous farce." Popular adhesion remained superficial.

Phase 2 - Cult of the Supreme Being (1794): Robespierre, at the height of his power, established a deist state religion. It acknowledged the existence of a Supreme Being and the immortality of the soul. He designed festivals, rituals, ceremonies. On June 8, 1794, he

presided over the Festival of the Supreme Being at the Champ de Mars: artificial mountain, processions, collective oaths.

Voltaire's quote that Robespierre repeated: "Si Dieu n'existait pas, il faudrait l'inventer" (If God did not exist, it would be necessary to invent him).

Result: Robespierre was guillotined six weeks later (July 28, 1794). But his diagnosis was correct: pure reason did not suffice.

4.3 Case 2: Auguste Comte and the Religion of Humanity (1849)

The founder of positivism, the philosopher who most explicitly rejected metaphysics, ended up creating a religion.

Phase 1 - Scientific positivism: Comte developed the "law of the three stages": theological, metaphysical, positive. Humanity progresses by abandoning supernatural explanations and adopting scientific method. Only observable, verifiable phenomena deserve attention.

The recognized problem: Comte understood that society requires cohesion. Traditional religion fulfilled that function. Without a replacement, scientific society would fragment.

Phase 2 - Religion of Humanity (1849): Comte designed a complete secular religion: - Object of worship: Humanity (Grand-Être) - Priesthood: sociologist-priests who would also be physicians - Seven sacraments (presentation, initiation, admission, destination, marriage, maturity, transformation, incorporation) - Calendar of secular "saints": each day dedicated to a thinker or hero of humanity - Rituals, prayers, festivals

Thomas Huxley described it as "Catholicism minus Christianity." John Stuart Mill, who initially admired Comte, said the ritual details "could have been written by no man who had ever laughed."

Result: The religion was a practical failure. Followers remained few. But Comte's diagnosis was correct: rational content needed an emotional vehicle.

4.4 Case 3: The Soviet Union (1917-1953)

Dialectical materialism, the 20th century's most explicitly anti-religious ideology, developed all the elements of a political religion.

Phase 1 - Militant atheism: Marx: "Religion is the opium of the people." The Soviet state actively persecuted religion, closed churches, confiscated ecclesiastical properties. The League of Militant Atheists (1925) had five million members by 1932.

Phase 2 - Political religion: - **Lenin's Mausoleum:** The embalmed body displayed like a saint's relic. Soviets pilgrimaged to view it. The Mausoleum became the liturgical center of Red Square. - **"Red Corner":** Traditional Russian homes had a corner for religious icons. After the revolution, icons were replaced by photos of Lenin and Stalin. The ritual structure remained; content changed. - **Festivals and liturgy:** Parades in Red Square, collective

oaths, songs of praise. Leaders observed from Lenin's Mausoleum like clergy from an altar. - **Saints and martyrs:** Fallen Bolsheviks, socialist work heroes. Statues, commemorations, veneration. - **Sacred texts:** Works of Marx-Engels-Lenin-Stalin treated as scripture. Citing them was obligatory; contradicting them, heresy.

A 1920 Soviet poster was titled "The Ten Commandments of the Proletariat."

Result: The system lasted 74 years. Binding worked, though programmatic content eventually collapsed.

4.5 Case 4: Turkish Kemalism (1923-present)

Mustafa Kemal Atatürk implemented the Islamic world's most aggressive modernization program. Radical secularism, Latin alphabet, Western dress, abolition of the caliphate.

Phase 1 - Positivist secularism: Turkey must become a secular republic in the French style. Religion was an obstacle to progress.

Phase 2 - Kemalist civil religion: - **Venerated figure:** "Atatürk" means "Father of the Turks." Official cult of the founder. - **Mandatory rituals:** Wreath-laying ceremonies at Atatürk monuments. Regulation 28283 (2012) specifies: who presides, official order, wreath specifications, attendance obligation. - **Sacred days:** National holidays with prescribed liturgy. - **Sacralization of secularism:** Laicism itself became dogma. Questioning it was heresy (*irtica*).

Scholars have noted the paradox: Kemalism rejected metaphysical claims on positivist grounds, yet constructed an elaborate civil religion with rituals, sacred symbols, and quasi-theological commitments to secularism itself.

Result: Kemalism survived decades. Erdoğan is eroding it, but not through rational arguments, rather through another binding: political Islamism.

4.6 Case 6: Contemporary Latin America (2018-present)

The contemporary Latin American cases represent the most explicit and deliberate instances of content-vehicle separation combined with structural attacks on embedded networks.

4.6.1 Brazil (2018-2022): Conservative-evangelical hybridization

The Brazilian conservative's trajectory illustrates the pattern with particular clarity because he combined Chicago School economic orthodoxy (Paulo Guedes as Economy Minister) with explicit evangelical mobilization.

Phase 1 - Economic program (low binding): - Pension reform (2019): Raised minimum retirement age, reduced benefits - Fiscal ceiling maintenance: Constitutional amendment limiting spending growth - Privatization agenda: Eletrobras, postal service, airports - Labor reform: Flexibilization, reduction of union power - Deregulation: Over 3,500 normative acts revoked in first two years

This program, designed by Guedes (PhD University of Chicago), represented orthodox liberalism indistinguishable from 1990s Washington Consensus prescriptions. Pure content, minimal binding.

Phase 2 - Transmission vehicle (high binding): - Explicit evangelical identity: Public baptism in Jordan River (2016), presided over by evangelical pastor Everaldo Dias Pereira - Religious symbolism: Campaign slogan "Brasil acima de tudo, Deus acima de todos" (Brazil above all, God above everyone) - Institutional alliance: Support from Assemblies of God (12 million members), Baptist Convention, Universal Church of the Kingdom of God - Congressional evangelicals: Evangelical bloc (Frente Parlamentar Evangélica) comprised over 30% of Chamber of Deputies (2018-2022) - Culture war framing: "Good citizens" (evangelical, traditional family) versus "cultural Marxism" (universities, artists, PT networks)

Phase 3 - Structural attack on embedded networks: - Anti-PT rhetoric: Workers' Party networks in unions, universities, social movements, cultural institutions framed as "o sistema" - Operation Car Wash exploitation: Corruption investigations delegitimized leftist establishment - Media confrontation: Attacked mainstream press (Globo, Folha) as "fake news" - Union weakening: Labor reform specifically targeted union financing (end of mandatory contributions) - Public sector cuts: Federal employee hiring freeze, ministry consolidation from 29 to 22

Outcomes and binding persistence: - 2018 election: 55.1% runoff despite radical program and minimal campaign resources - Economic crisis 2020-2021: COVID-19 recession, 4.1% GDP contraction (2020), inflation rising to 10.06% (2021) - Approval trajectory: Despite economic deterioration, maintained 30-35% "great/good" approval (Datafolha polls 2020-2022), floor never fell below 25% - 2022 reelection attempt: Narrowly defeated (49.1% versus Lula 50.9%), but maintained 58 million votes despite recession, COVID mismanagement, corruption scandals - Counterfactual: Temer government (2016-2018), similar liberal economic program without religious binding, ended with 3% approval

The Brazilian case demonstrates binding resilience to instrumental failure. Orthodox economic program produced minimal growth (1.2% average annual GDP growth 2019-2022, excluding pandemic year). Yet electoral floor remained structurally higher than previous liberal governments lacking religious binding.

4.6.2 Argentina (2023-present): Multi-confessional hybridization

The Argentine case is analyzed in detail in Section V. Key distinction from Brazilian conservative (2018-2022): even more explicit diversification across multiple binding sources (Judaism, evangelicalism, Catholicism) and more aggressive structural attacks ("motosierra" rhetoric, immediate shock therapy).

4.6.3 El Salvador: Nayib Bukele (2019-present)

Bukele represents a variant where binding derives less from explicit religious identity and more from charismatic personalism and national refoundation narrative.

- Economic program: Fiscal orthodoxy, Bitcoin adoption (2021), foreign investment attraction
- Binding source: "Bukele-ismo" personality cult, millennial aesthetic, "coolest dictator" branding, explicit messianismism ("I am the instrument, God is the force")
- Structural attack: Dismantling traditional parties (FMLN left, ARENA right), attacking judiciary, purging attorney general, concentrating power
- Electoral outcome: 85% approval sustained over four years, reelection (2024) with over 80% despite constitutional prohibition
- Binding persistence: Maintains approval despite concerns about authoritarianism, press freedom deterioration, judicial independence erosion

What unites these three cases is not ideological content (the Brazilian conservative and the hybrid liberal lean libertarian, Bukele more populist) but structural pattern: economic program + high-binding vehicle + attack on embedded networks. The binding source varies (evangelicalism, personality cult, nationalist refoundation), but function remains constant: generating identity adhesion independent of instrumental outcomes while delegitimizing alternative power centers.

4.7 Comparative table

Case	Place/Era	Rational project	Adopted binding	Revealing quote
French Revolution	France 1789-94	Cult of Reason (atheism)	Cult of Supreme Being	"If God did not exist, it would be necessary to invent him" (Voltaire/Robespierre)
Positivism	France 1849	Science as sole knowledge	Religion of Humanity	"Catholicism minus Christianity" (Huxley on Comte)
USSR	Russia 1917-53	Dialectical materialism	Cult of Lenin/Stalin, mausoleum, rituals	"The Ten Commandments of the Proletariat" (1920 poster)
Kemalism	Turkey 1923-	Positivist secularization	Civil religion with venerated Atatürk figure	"The ideology is almost entirely a religious organization" (*Turkish Studies* 2025)
Argentine Enlightenment	Argentina 1821-27	Rivadavian project	Romanticism (Generation of 37)	"Fatal is the illusion... that pretends to change the nature of things" (Rivadavia)
LatAm Liberalism	Argentina/Brazil 2018-	Austrian/Chicago orthodoxy	Evangelicalism, religious symbolism	"Political outsiders betting on faith as fuel" (paraphrase)

4.8 The anthropological constant

Six cases. Two centuries. Three continents. Opposed programs: Enlightenment republicanism, positivism, communism, Islamic secularism, classical liberalism. The only commonality is being rationalist projects. And all faced the same problem.

This suggests the binding deficit is not a specific flaw of liberalism (as Mises believed) but a property of any program relying exclusively on reason for propagation. It is an anthropological constant, not a historical accident.

Jonathan Haidt, from moral psychology, reached the same conclusion by another route: humans are not primarily rational. They have "moral foundations" including loyalty, authority, sanctity. Programs appealing only to reason ignore most of human psychological architecture.

From the EPT framework: reason is a possible component of memetic content, but not an efficient transmission vehicle. Memes that replicate best activate multiple psychological channels, not only the cognitive.

V. THE CONTEMPORARY HYBRIDIZATION

5.1 The phenomenon observed

Leaders combining liberal economic programs with high-binding elements:

Case	Country	Economic program	Binding elements
Argentine libertarian (2023-)	Argentina	Austrian orthodoxy	Multi-confessional religious outreach; religious symbolism
Brazilian conservative (2018-2022)	Brazil	Chicago liberalism	Explicit evangelicalism, military nationalism, religious symbolism
Salvadoran populist (2019-)	El Salvador	Fiscal orthodoxy + Bitcoin	Charismatic imagery, national refoundation narrative

5.2 Evidence of deliberation

The Argentine case offers indications that hybridization is not accidental:

- **March 2024:** the hybrid liberal participates in Purim ceremony with Orthodox Jewish community
- **June 2024:** Public meeting with evangelical leaders at Luna Park stadium
- **July 22, 2025:** Decree 486/2025 published in Boletín Oficial, granting legal personhood to non-Catholic religious organizations nationwide (https://www.boletinoficial.gob.ar/detalleAviso/primera/328661/20250722)
- **Christmas Eve 2024:** Presidential Grenadier Guards transfer image of Baby Jesus to nativity scene in official ceremony

Diversification across different religious traditions suggests strategy, not genuine conversion. The objective appears to be capturing binding from multiple sources without exclusive commitment to any.

5.3 Memetic logic

Hybridization resolves Mises's dilemma by separating components:

- **Programmatic core** (low binding, high coherence): free markets, private property, deregulation
- **Transmission vehicle** (high binding, low coherence): religious narratives, ritualism, tribal belonging

Content travels in a container not its own. From a pure rationalist perspective, this is incoherent. From an evolutionary perspective, it is adaptive.

5.4 The inverted vector: Religious leaders entering politics

If liberal politicians represent rationalists seeking mystical fuel, evangelical pastors entering politics represent the inverse phenomenon: providers of mystical fuel entering the political arena directly. The emergence of pastoral presidential candidacies, pushed by traditional union sectors, reveals that the "discovery" of binding as a scarce resource has generalized across the political spectrum.

Juan Pablo Brey, secretary-general of Aeronavegantes and CGT leader, articulates the hypothesis unknowingly:

> "Estamos pasando una crisis que es económica, pero también social. Una crisis de valores, de la política, de liderazgo. Y todo lo que vemos hoy no son propuestas sino peleas personales entre dirigentes. Entonces, somos varios los que creemos que una figura como la de the evangelical pastor, un divulgador de valores, es lo que el país necesita." (Infobae, December 2, 2024)

Translation: We are experiencing a crisis that is economic but also social. A crisis of values, of politics, of leadership. And all we see today are not proposals but personal fights between leaders. So, several of us believe that a figure like the evangelical pastor, a disseminator of values, is what the country needs.

The diagnosis is explicit: "traditional politicians do not generate the expectation that Dante generates." Programmatic content (Peronism, defense of labor) needs a new vehicle because the previous one is exhausted.

Letra P formulates it with journalistic clarity: "Para un país que ya votó a un outsider místico que habla con sus perros muertos, el desembarco de un pastor carismático con estética de estrella pop y mensaje motivacional no suena tan extraño. Los outsiders políticos compiten por el electorado religioso." (Translation: For a country that already voted for a mystical outsider who talks with his dead dogs, the arrival of a charismatic pastor with pop-star

aesthetic and motivational message does not sound so strange. If the hybrid liberal found in fury his political fuel, the pastor bets on faith.)

Quantitative evidence of religious binding as contested resource:

1. **Evangelical demographic expansion:**
   - 2008: ~10% of Argentine population (Proyecto ARDA estimates)
   - 2024: ~20% (Aciera calculations, Revista Noticias August 2025)
   - Growth rate: 100% increase over 15 years, versus Catholic Church decline from 90% (1960) to 62.9% (2019 census)
   - Absolute numbers: ~9 million evangelicals in population of 45 million
2. **Electoral institutionalization:**
   - 2023: First confessional evangelical legislative bloc in Congress
   - 2023 runoff: the hybrid liberal 55.7%, Massa 44.3% (National Electoral Chamber official data)
   - 2023 PASO: La Libertad Avanza obtained 30.04% versus PRO's 28.27% (pre-runoff)
   - Key difference: The administration's religious hybridization versus Macri/PRO's secular liberalism
3. **Political competition for evangelical binding:**
   - CB Consultora poll (December 2025): the evangelical pastor registered 1.8% voting intention as first-time measured candidate (Perfil, December 15, 2025)
   - The pastor's appearance in polls follows public declarations by CGT leaders (Brey, Aeronavegantes) promoting candidacy
   - Cross-ideological convergence: The liberal right and Peronist unionists competing for same evangelical constituency
4. **Institutional formalization of religious links:**
   - Decree 486/2025 (July 22, 2025): Grants legal personhood to non-Catholic religious organizations nationwide
   - Eliminates requirement for evangelical churches to register as "civil associations"
   - Allows direct property ownership, autonomous governance, and fiscal benefits
   - Practical effect: Reduces bureaucratic friction for evangelical institution-building
5. **Temporal coincidence: Binding + Structural Attack:**
   - March 2024: the libertarian's Purim event with Orthodox Jewish community
   - June 2024: Luna Park evangelical meeting (attendance estimated 15,000)
   - July 2025: Decree 486/2025 published
   - Simultaneous period: Deregulation Decree 70/2023 (December 2023), public sector cuts (20,000+ state employees dismissed Q1 2024), confrontations with CGT (general strikes March and May 2024)

This convergence pattern confirms the scarcity hypothesis: binding is a limited resource that political actors from opposed ideological positions recognize as valuable and compete to capture. The simultaneity of The administration's religious outreach and structural attacks on embedded networks (unions, state corporations) supports the dual-dimension hypothesis.

V.6 COMPARATIVE ANALYSIS: NATURAL EXPERIMENTS IN BINDING STRATEGIES

The theoretical framework gains empirical power through comparative analysis of leaders with similar economic programs but differing binding strategies. Argentina provides a natural experiment: Mauricio Macri (2015-2019) and Argentine libertarian administration (2023-present) both pursued liberal economic reforms in the same national context, electoral system, and constitutional framework. The key difference: binding coefficient.

5.6.1 Argentina: The technocratic-hybrid natural experiment

Macri (2015-2019): Liberal economics without binding

Mauricio Macri's PRO (Propuesta Republicana) represented technocratic liberalism: graduate of Universidad Católica Argentina, businessman (Socma Group), president of Boca Juniors football club (1995-2007). His 2015 campaign emphasized managerial competence, institutional normalization, and gradual economic reform.

Economic program (content): - Elimination of currency controls (December 2015) - Reduction of export taxes on agriculture (gradual phase-out) - Fiscal consolidation (reducing primary deficit from 4.8% GDP in 2015 to 2.7% in 2018) - Central Bank independence - Labor market flexibilization proposals - Pension reform (2017) - Tax reform (2017) - Debt market reintegration (2016)

This program was orthodox liberalism, indistinguishable from Washington Consensus prescriptions.

Transmission vehicle (binding): - $\beta \approx 0.25$ (estimated): PRO functioned as management party, not identity community - Campaign slogan: "Cambiemos" (Let's change) - instrumental framing, not identity marker - No religious elements: Macri personally Catholic but no mobilization of religious communities - No ritualistic elements: Campaign events were rallies, not ceremonies - Business background framed as competence credential, not tribal belonging - Urban professional aesthetic: suit-and-tie, moderate language, technocratic presentation - Coalition strategy: Broad coalition including centrists, disaffected Peronists, anti-Kirchnerists

Structural attack on embedded networks: - Limited: Confrontations with unions (general strikes in 2016, 2017, 2018, 2019) but no delegitimizing rhetoric - No "casta" framing: Macri portrayed opposition as mistaken, not corrupt - Inherited state structure largely maintained

- Public employment grew 3.5% during term (208,000 new state employees) - Major state enterprises (YPF, Aerolíneas Argentinas) not privatized - Subsidies reduced gradually but not eliminated

Electoral and approval outcomes: - 2015 runoff: 51.4% (narrow victory, required coalition with Radical Party) - Approval trajectory: Started 50% (December 2015), declined to 35% (December 2016), stabilized 35-40% (2017), collapsed to 25-30% (2018-2019) during economic crisis - 2019 reelection attempt: 40.4% (lost to Fernández-Fernández ticket) - Economic crisis response: Approval fell from 38% (March 2018) to 28% (September 2018) following currency crisis - **Key observation:** No approval floor. Support declined linearly with economic deterioration.

Why Macri failed (binding deficit perspective):

Macri's coalition was purely instrumental. Voters supported him to achieve outcomes (lower inflation, economic growth, institutional normalization). When outcomes failed to materialize (2018 inflation 47.6%, 2019 recession 2.1% GDP contraction, currency depreciation 50% in 2018), support evaporated.

Low binding ($\beta \approx 0.25$) meant: - No identity adhesion independent of results - No resilience to economic shocks - No volunteer mobilization capacity during crises - No organic transmission through communities - Coalition collapse under stress (Radical Party, Peronist defectors abandoned ship)

Additionally, limited structural attack meant embedded networks maintained resistance capacity. Unions retained membership, financing (despite labor reform attempts), and mobilization power. State corporations preserved internal solidarity. "Kirchnerist" identity remained strong among ~30% of electorate, providing alternative binding for opposition.

Argentine hybrid administration (2023-present): Liberal economics with evangelical binding

Javier The Argentine libertarian leader's background: academic economist (Universidad de Belgrano, ESEADE), libertarian ideologue, TV panelist (2016-2021), congressman (2021-2023), president (2023-present). Campaign emphasized radical rupture, not gradual reform.

Economic program (content): - More radical than Macri but same ideological family: - Dollarization proposal (not implemented but signaled commitment) - Central Bank abolition proposal (not implemented but rhetorically powerful) - Deregulation Decree 70/2023 (December 2023): 366 articles deregulating economy - Fiscal shock therapy: 3% GDP primary surplus target 2024 (versus 2.3% deficit 2023) - Public employment cuts: 20,000+ state employees dismissed Q1 2024 - Subsidy elimination: Energy, transport subsidies reduced drastically - Privatization agenda: Aerolíneas Argentinas, Radio y Televisión Argentina, others proposed

This program is more extreme than Macri's but same direction: less state, more market, fiscal discipline, deregulation.

Transmission vehicle (binding): - β ≈ 0.55 (estimated, rising): Hybrid binding under construction - Religious elements: - Purim ceremony (March 2024) with Orthodox Jewish community - Luna Park evangelical meeting (June 2024, 15,000+ attendees) - Decree 486/2025 (July 2025) legal personhood for non-Catholic organizations - Christmas nativity ceremony (December 2024) - Charismatic aesthetics: "Sent by God" rhetoric, prophetic framing, divine mission language - Ritualistic campaign events: Chainsaw symbol, lion imagery, "las fuerzas del cielo" (forces of heaven) framing - Personal narrative: Economic suffering, struggle against "casta," redemption arc - Tribal identity markers: "libertarios" versus "estatistas," "alphas" versus "betas," in-group language - Rock concert aesthetic: Campaign rallies resembled concerts, not political meetings

Structural attack on embedded networks: - Aggressive and explicit: - "La casta" framing: Political class as parasitic, corrupt, enemies of the people - Union confrontations: CGT general strikes (January, March, May 2024) but the administration doubled down rhetorically - Media attacks: Traditional press framed as propaganda arm of establishment - State apparatus purge: Ministry consolidation (18 to 8), dismissals without compensation - Privatization rhetoric: "Todo lo que puede estar en manos privadas, debe estar en manos privadas"

Electoral and approval outcomes (preliminary, term ongoing): - 2023 PASO: 30.04% (surprising first place) - 2023 first round: 29.98% (second place, 5 points behind Massa) - 2023 runoff: 55.7% (decisive victory, +11.4 points versus first round) - Initial approval: 50-55% (December 2023-January 2024) - Economic shock phase: Approval 45-50% (February-June 2024) despite inflation spike (25.5% monthly December 2023) - Stabilization phase: Approval 50-55% (July-December 2024) as monthly inflation declines - **Key observation:** Approval maintained despite unprecedented shock (December 2023 inflation 25.5%, annual 2023 inflation 211.4%). No collapse to Macri-level lows.

Why the hybrid approach (so far) succeeds where Macri failed:

The hybridized administration's support base includes significant identity-based component. Evangelical communities (20% population), libertarian ideologues, "anti-casta" tribal identity provide binding independent of immediate economic outcomes.

Higher binding (β ≈ 0.55, estimated) means: - Resilience to economic shocks: Supporters interpret short-term pain as necessary medicine, not failure - Volunteer mobilization: Organic social media defense, grassroots organization without party infrastructure - Generational transmission: Families identify as "libertarios," not merely libertarian voters - Ritualistic participation: Campaign events, online communities, symbolic consumption (chainsaw memes, lion imagery)

Additionally, aggressive structural attack delegitimizes resistance. "La casta" framing portrays union strikes not as legitimate protest but as corrupt interests defending privilege. This asymmetry weakens embedded networks' moral authority even as their material power persists.

Controlled comparison (same country, different β):

Variable	Macri (2015-2019)	Hybrid liberal (2023-)
Economic program ideology	Liberal orthodox	Liberal radical
Binding coefficient (β)	≈ 0.25	≈ 0.55
Religious mobilization	None	Explicit (multi-confessional)
Charismatic framing	None	Central
"Casta" delegitimization	Weak	Aggressive
Initial approval	50%	50-55%
Approval under crisis	25-30% (collapsed)	45-50% (sustained)
Electoral outcome	Lost reelection	(TBD 2027)

The natural experiment is imperfect (different economic contexts, different opposition, COVID-19 intervening variable for Macri), but the pattern is consistent with binding hypothesis: higher β predicts greater resilience to instrumental failures.

5.6.2 Brazil: Competing high-binding movements

Brazil offers complementary comparison: two leaders with high binding coefficients but opposed sources, competing for same electorate.

Brazilian conservative administration (2018-2022): Evangelical + military nationalism

Binding sources: - Evangelical identity (explicit): Baptism in Jordan River (2016), campaign slogan "Brasil acima de tudo, Deus acima de todos" - Military nationalism: Army captain background, pro-dictatorship rhetoric (1964-1985 regime portrayed positively), law-and-order framing - Anti-PT tribalism: Workers' Party framed as communist threat, moral corruption, civilizational enemy - Traditional family values: Opposition to abortion, LGBTQ+ rights, gender ideology in schools

Estimated β ≈ 0.60: Strong identity component. Brazilian conservative supporters identified as evangelical Christians, patriots, defenders of family, anti-communists. Support transcended policy performance.

Electoral outcomes: - 2018 runoff: 55.1% (despite minimal party infrastructure, limited campaign time) - 2022 runoff: 49.1% (lost but maintained 58 million votes despite recession, COVID-19 mismanagement, corruption scandals)

Approval resilience: - Never fell below 25% even during worst crises (April 2021: 24%, but 30% "great/good" among evangelicals) - Evangelical bloc maintained support: 65% approval among evangelicals versus 30% general population (Datafolha, March 2021)

Luiz Inácio Lula da Silva (1980-present): Union + social movement binding

Binding sources: - Working-class identity: Metalworkers union leader, nordestino migrant narrative, "filho do Brasil" (son of Brazil) framing - Social movement history: Strikes in 1970s-1980s against dictatorship, founding of PT (Workers' Party, 1980) - Poverty-to-presidency

narrative: Personal story of overcoming, literacy at age 10, factory worker to president - Redistributive mystique: Bolsa Família as sacred program, lifting 20+ million from poverty (2003-2010)

Estimated β ≈ 0.70: Very high identity adhesion. Lula supporters (especially nordeste, working class, Afro-Brazilians) identify as "petistas" across generations. Support survives imprisonment (2018-2019), corruption scandals, policy failures.

Electoral persistence: - Presidential candidate six times (1989, 1994, 1998, 2002, 2006, 2022) - Despite imprisonment (2018-2019), returned to win 2022 (50.9%) - Approval during first term (2003-2006) remained above 50% even when economy underperformed

Comparative dynamics:

2022 election represented clash of bindings: - Brazilian conservative (2018-2022): evangelical + military + anti-PT (β ≈ 0.60) - Lula: working-class + social movements + nordeste (β ≈ 0.70)

Result: 50.9% Lula, 49.1% the Brazilian conservative (1.8 percentage point margin, 2.1 million votes)

Both leaders maintained massive support despite instrumental failures: - Brazilian conservative (2018-2022): 360,000 COVID-19 deaths, vaccine denialism, economic recession, corruption scandals → still 49.1% - Lula: Operation Car Wash corruption revelations, impeachment of successor (Dilma), imprisonment → still 50.9%

Binding hypothesis explanation: When β is high for both competitors, electoral outcome depends on relative binding strength and demographic distribution of binding sources. Lula's β ≈ 0.70 (PT's 40-year organizational history, deeper community roots) slightly exceeded the Brazilian conservative's β ≈ 0.60 (evangelical mobilization only since 2016-2018).

Counterfactual: Had PT nominated low-binding technocrat (e.g., Fernando Haddad in 2018, received 44.9%), the Brazilian conservative's binding advantage would have been decisive. Lula's personal binding compensated for the Brazilian evangelical mobilization.

Lesson for liberal reformers: In contexts where opposition possesses high binding, liberal reform requires competitive binding, not merely superior policy. Macri faced Kirchnerism (β ≈ 0.65) with PRO (β ≈ 0.25) and lost. the hybrid administration faces Kirchnerism with hybrid binding (β ≈ 0.55, rising) and sustains approval despite harsher policies. the Brazilian conservative faced Lula (β ≈ 0.70) with evangelical binding (β ≈ 0.60) and came within 1.8 points.

5.6.3 Cross-regional patterns: Modi, Duterte, Orbán

The pattern extends beyond Latin America. Three cases from distinct regions demonstrate universality:

Narendra Modi (India, 2014-present): Hindu nationalism + economic liberalization

- Economic program: "Make in India" industrialization, foreign investment attraction, labor law reforms, GST tax reform
- Binding source: Hindu nationalism (Hindutva), BJP/RSS organizational networks, anti-Muslim framing, Ayodhya temple (2019)
- Estimated $\beta \approx 0.65$
- Electoral persistence: 2014 (31.0% vote share, 282 seats), 2019 (37.4%, 303 seats), 2024 (36.6%, 240 seats but coalition)
- Approval resilience: Maintained 60%+ approval despite demonetization crisis (2016), farmer protests (2020-2021), COVID-19 mismanagement

Rodrigo Duterte (Philippines, 2016-2022): Catholic populism + strongman persona

- Economic program: Infrastructure spending ("Build Build Build"), fiscal consolidation, foreign investment
- Binding source: Catholic religiosity (despite Vatican criticism), Davao strongman legend, anti-drug crusade framed as moral purification
- Estimated $\beta \approx 0.55$
- Electoral outcome: 39% plurality (2016, six-way race)
- Approval persistence: 70-80% approval sustained throughout term despite extrajudicial killings, ICC investigation, pandemic mismanagement

Viktor Orbán (Hungary, 2010-present): Christian nationalism + illiberalism

- Economic program: State capitalism, EU fund absorption, industrial policy, family subsidies
- Binding source: Christian Europe defense, anti-immigration framing, Hungarian nationalism, civilizational struggle rhetoric
- Estimated $\beta \approx 0.60$
- Electoral dominance: 2010 (52.7%), 2014 (44.9%), 2018 (49.3%), 2022 (54.1%)
- Approval resilience: Maintained majority despite corruption allegations, EU conflicts, inflation crises

All three cases combine: 1. Economic programs (varying content but all involve market elements + state intervention) 2. High-binding cultural/religious vehicles (Hindu nationalism, Catholic populism, Christian nationalism) 3. Structural attacks on opposition networks (secular institutions, liberal NGOs, opposition media)

The pattern transcends ideology: Modi right-wing, Duterte center-populist, Orbán illiberal. What unites them is high β + structural attack formula.

5.6.4 Theoretical implications of comparative analysis

The cross-case comparisons support five theoretical propositions:

Proposition 1 (Binding advantage): Leaders with higher β outperform leaders with similar economic programs but lower β in competitive elections, controlling for economic context.

Evidence: Argentine hybrid > Macri (Argentina), the Brazilian conservative ≈ Lula (Brazil, Lula's slight β advantage yielded narrow victory), Modi's sustained dominance despite policy failures.

Proposition 2 (Approval floor): High-β leaders exhibit higher approval floors during economic crises than low-β leaders facing similar crises.

Evidence: The hybrid administration maintained 45-50% during shock (2024) versus Macri's collapse to 25-30% (2018-2019). The Brazilian conservative never below 24% despite COVID disaster. Duterte sustained 70%+ despite extrajudicial killings.

Proposition 3 (Competitive binding): When both competitors possess high β, outcome depends on relative β-values and demographic distribution of binding sources.

Evidence: Lula (β ≈ 0.70) narrowly defeated Brazilian conservative (β ≈ 0.60). Modi's sustained dominance reflects BJP/RSS organizational β versus fragmented opposition.

Proposition 4 (Necessary but insufficient): High β enables resilience but does not guarantee policy implementation. Structural attacks on embedded networks are independently necessary.

Evidence: Duterte maintained approval but failed to implement economic reforms (unions and oligarchs blocked). The administration's aggressive "casta" attacks may enable implementation where the Brazilian conservative's partial attacks failed.

Proposition 5 (Source fungibility): Binding sources are partially fungible. Religious, nationalist, ethnic, class-based identities can substitute for each other as vehicles for policy content.

Evidence: Argentine libertarian (multi-confessional religious), Brazilian conservative (evangelical), Modi (Hindu), Duterte (Catholic), Orbán (Christian Europe), Lula (class/movement). Different sources, similar binding function.

These propositions, if confirmed through systematic testing, suggest binding coefficient is fundamental variable in political economy, comparable in importance to traditional factors (institutions, economic performance, party organization).

V.5 METHODOLOGICAL NOTE: MEASURING BINDING AND TESTING PREDICTIONS

Before presenting testable predictions, methodological transparency requires specifying how binding coefficients can be operationalized and how predictions can be falsified.

5.5.1 Operationalizing the binding coefficient

The binding coefficient β, defined as $\beta = \frac{\text{Identity adhesion}}{\text{Identity adhesion + Instrumental expectations}}$, requires measurement instruments that distinguish identity-based from outcome-based motivations.

Survey-based approach:

Standard political preference surveys confound these dimensions. A respondent saying "I support candidate X" may do so because (a) X represents my identity/community, (b) I expect X to deliver beneficial outcomes, or (c) both. Existing surveys (World Values Survey, LAPOP) do not systematically distinguish.

Proposed instrument (Appendix A provides full version):

1. **Direct identity question:** "Regardless of whether [party/movement] delivers on its promises, do you feel it represents who you are as a person?"
   - Strongly agree (high β) → Weakly disagree (low β), 7-point scale
2. **Conditional support question:** "If [party/movement]'s policies produced economic deterioration, would you continue supporting it?"
   - Definitely yes (high β) → Definitely no (low β), 7-point scale
3. **Community embeddedness question:** "How many people in your family/close social circle share your political preference?"
   - Almost all (high β) → Almost none (low β), 5-point scale
4. **Transmission question:** "Did you adopt this political preference mainly through (a) family/community tradition, or (b) independent evaluation of policy proposals?"
   - Family/tradition (high β) → Independent evaluation (low β), binary with intensity scale

Scoring: β = (Identity items sum) / (Identity items sum + Instrumental items sum), normalized to 0-1 range.

Validation requirements: Instrument must demonstrate: - Construct validity: Correlates with theoretically predicted behaviors (party switching, protest participation, resilience to negative information) - Discriminant validity: Distinguishes binding from mere policy agreement - Test-retest reliability: Stable across time periods absent major shocks - Cross-cultural validity: Functions in WEIRD and non-WEIRD contexts

Behavioral proxies when survey data unavailable:

In contexts where direct measurement is infeasible, proxy indicators include: - **Electoral volatility:** Low-binding movements should show high voter switching rates - **Protest turnout resilience:** High-binding movements should maintain mobilization capacity despite instrumental failures - **Generational transmission:** High-binding preferences should show parent-child correlation independent of outcomes - **Content analysis:**

Frequency of identity markers ("we," "our people," "traitors") versus instrumental claims ("growth," "efficiency," "evidence") in political communication

5.5.2 Coding political communication for binding elements

Systematic content analysis of speeches, social media, campaign materials requires codebooks specifying binding indicators:

Religious/sacred binding: - References to deity, scripture, religious community - Participation in religious ceremonies as political acts - Endorsements from religious authorities - Religious metaphors and framing

Tribal/ethnic binding: - In-group/out-group language ("we the people" versus "elites," "the caste") - Ethnic or regional identity markers - References to shared history, ancestors, tradition - Boundary maintenance rhetoric (who belongs, who does not)

Nationalist binding: - Sovereignty narratives, foreign threats - National symbols (flags, anthems, heroes) - Refoundation language ("new birth," "return to greatness") - Territorial or historical grievances

Instrumental (non-binding) content: - Policy specifics, numerical targets - Cost-benefit analyses, technical arguments - Evidence citations, expert appeals - Conditional promises ("if we do X, Y will result")

Coding procedure: Random sample of political communication texts, dual coders, Cohen's kappa for intercoder reliability (target ≥ 0.70), proportion of binding versus instrumental content calculated per text and aggregated per leader/movement.

5.5.3 Establishing causality versus correlation

The central challenge: Does hybridization cause electoral success, or do electorally successful leaders happen to have religious predispositions?

Strategies for causal inference:

1. **Temporal sequencing:** Document whether religious outreach precedes or follows electoral success. The administration's religious events (March-June 2024) occurred *after* electoral victory (November 2023) but *before* policy results were measurable, suggesting strategic binding-building rather than victory celebration.

2. **Counterfactual comparison:** Compare leaders with similar economic programs but differing binding strategies within same electoral context. Argentina 2015-2019 (Macri: liberal economics, no religious binding) versus 2023-present (the hybrid liberal: liberal economics, explicit religious binding) provides natural experiment.

3. **Cross-national replication:** If binding effect is causal, pattern should replicate across different national contexts. Brazil (Brazilian conservative), El Salvador (Bukele), Philippines (Duterte's Catholic nationalism), India (Modi's Hindu

nationalism) provide variation in economic programs but common binding + structural attack pattern.

4. **Mechanism tracing:** If binding operates through identity adhesion independent of outcomes, should observe (a) approval floor during economic crises higher for high-β than low-β leaders, (b) supporter retention despite policy reversals higher for high-β movements, (c) intensity of support (protest turnout, donations, volunteer time) higher for high-β movements controlling for policy agreement.

5.5.4 Falsification criteria

Scientific predictions must specify observable outcomes that would falsify the theory.

Binding deficit hypothesis falsified if: - Pure rationalist programs consistently outperform hybridized variants in mass elections under conditions of restored memetic competition - Approval trajectories during economic crises show no difference between high-β and low-β leaders controlling for policy outcomes - Content analysis reveals no systematic difference in identity versus instrumental messaging between successful and unsuccessful reform movements - Historical cases show rationalist projects sustaining mass mobilization for extended periods without incorporating high-binding elements

Dual-dimension hypothesis (binding + structural attack) falsified if: - High-binding movements without structural attacks show similar reform success rates as those combining both dimensions - Structural attacks without binding mechanisms show similar electoral survival rates as dual-strategy cases - Binding elements predict success but attacks on embedded networks do not add explanatory power

Content-vehicle separation hypothesis falsified if: - Hybridized movements show policy implementation rates indistinguishable from pure movements (suggesting vehicle does not protect content) - Internal tensions between rationalist and mystical components systematically destroy movements before policy implementation (suggesting separation is unstable) - Cross-spectrum competition for same binding sources does not occur, or occurs but fails to generate similar electoral outcomes (suggesting binding is not actually a scarce, valuable resource)

With methodological foundations established, we now present specific testable predictions.

VI. TESTABLE PREDICTIONS

6.1 Prediction 1: Electoral advantage of hybrids over purists

Hypothesis: In contexts of high memetic competition, hybridized liberal candidates will obtain better electoral results than pure liberals (without binding components).

Operationalization: Compare electoral performance of liberal candidates according to presence/absence of religious or nationalist elements in their communication.

Preliminary test: In Argentina's 2023 election, the hybridized liberal candidate obtained 55.7% in the runoff, while secular-liberal candidates from the traditional center-right coalition obtained lower vote shares in primaries.

Falsification criterion: If pure liberals systematically outperform hybrids in competitive environments, the hypothesis is falsified.

6.2 Prediction 2: Resilience to economic crises

Hypothesis: Hybridized liberal movements will show less support loss amid economic deterioration than low-binding movements.

Operationalization: Measure correlation between negative economic indicators and support loss in polls, controlling for estimated binding coefficient.

Future test: If the Argentine economy does not substantially improve in 2025-2026, theory predicts The hybridized administration's support will fall less than it would for a liberal government without binding components.

Measurement: Compare approval rating trajectories of the high-binding liberal versus Macri 2015-2019 (low β) under similar economic conditions.

Falsification criterion: If high-binding and low-binding leaders show identical response patterns to economic shocks, the binding coefficient is irrelevant.

6.3 Prediction 3: Observable internal tensions

Hypothesis: Hybridization generates tensions between rationalist and mystical components that will manifest in internal conflicts, desertions, or discursive inconsistencies.

Operationalization: Content analysis of public statements seeking contradictions between components; recording conflicts between "technical" and "ideological" factions.

Preliminary evidence: In Brazil, tension between Paulo Guedes (technical liberalism) and the Brazilian conservative's ideological wing was recurrent. In Argentina, criticism from LLA sectors of the pastor as an "artificial" project already shows predicted dynamics.

Falsification criterion: If hybridized movements show no more internal tension than ideologically homogeneous movements, the hypothesis is falsified.

6.4 Prediction 4: Competitive convergence

Hypothesis: Given the scarcity of religious binding as a resource, actors from different political spaces will compete to capture it, generating similar offers from opposed programs.

Preliminary evidence: The simultaneity of the libertarian right and sectors of Peronist unionism (traditional left) seeking the evangelical electorate confirms the prediction.

Falsification criterion: If religious binding remains confined to one ideological pole without cross-spectrum competition, the scarcity hypothesis is falsified.

6.5 Prediction 5: Dual-dimension strategy

Hypothesis: Successful liberal reformers will combine high-binding narratives with explicit attacks on entrenched power networks. Addressing only binding (without dismantling material interests) or only structural reforms (without binding) should show lower success rates.

Operationalization: Code political communication for (a) binding elements (religious references, tribal identity, ritualism) and (b) anti-establishment rhetoric targeting specific networks ("la casta," unions, state corporations, "o sistema").

Preliminary evidence: - **the hybrid liberal:** Evangelical binding (Purim, Luna Park, Decree 486/2025) coincides temporally with structural attacks (privatization, deregulation, public sector cuts, CGT confrontations) - **Brazilian conservative (2018-2022):** Religious symbolism parallels attacks on Workers' Party networks and public sector unions - **Counterfactual:** Macri (Argentina 2015-2019) attempted structural reforms without binding mechanisms and failed electorally; traditional Peronism maintains binding but avoids structural attacks on its own networks

Measurement: Compare policy implementation persistence and electoral survival across: - High-binding + structural attack (the hybrid liberal, the Brazilian conservative) - Low-binding + structural attack (Macri, traditional liberals) - High-binding + no structural attack (traditional Peronism, chavismo)

Falsification criterion: If success rates do not differ systematically across these categories, the dual-dimension hypothesis is falsified.

VII. IMPLICATIONS FOR EXTENDED PHENOTYPE THEORY

7.1 Binding as fifth dimension of lock-in

In prior work, I identified four mechanisms of constitutional lock-in: textual vagueness, treaty hierarchy, precedent weight, federal fragmentation. Binding constitutes a fifth mechanism, not technical-legal but psychological-identitarian, with real institutional effects.

High-binding institutions better resist reform attempts because their support base does not respond to cost-benefit analysis.

Formal extension: Constitutional Lock-In Index (CLI) should incorporate binding coefficient β:

$$CLI = f(\text{textual vagueness, treaty hierarchy, precedent, federal frag, } \beta)$$

where $\frac{\partial CLI}{\partial \beta} > 0$.

7.2 Revision of WEIRD/non-WEIRD gap and the substrate hypothesis

My prior finding of a 5.4% (WEIRD) versus 31.2% (non-WEIRD) constitutional implementation gap can be reinterpreted through the binding lens: institutions designed in WEIRD contexts assume a specific psychological substrate (instrumental rationality, trust in abstract rules, low-binding civic culture) that does not exist or competes with alternative substrates in non-WEIRD contexts.

The substrate hypothesis: Institutional fitness depends not only on functional design quality but on compatibility between the institution's assumed psychological substrate and the actual substrate in the implementation context.

Joseph Henrich's *The WEIRDest People in the World* (2020) documents how Western institutions coevolved with specific psychological adaptations: individualism, analytical thinking, impersonal trust, guilt-based morality. These adaptations emerged from particular historical circumstances (Catholic Church marriage policies breaking down clans, urbanization, literacy, market integration) not replicated elsewhere.

WEIRD institutions thus contain implicit binding assumptions: - **Legal personhood of corporations:** Assumes individuals can form identity adhesion to abstract entities beyond kinship - **Judicial independence:** Assumes judges' loyalty to abstract "rule of law" can override tribal, familial, or patronage ties - **Meritocratic civil service:** Assumes individuals evaluate state officials by performance rather than kinship/tribal belonging - **Anti-corruption frameworks:** Assume moral guilt from abstract norm violation motivates behavior more than in-group loyalty

These assumptions hold in low-binding (high-WEIRD) contexts. They fail in high-binding (low-WEIRD) contexts not because non-WEIRD people lack capacity but because alternative binding sources (kinship, tribe, patronage) generate stronger identity adhesion than abstract rules.

Empirical implications:

1. **Transplant failure rates should correlate with substrate mismatch:** A Swedish anti-corruption institution (designed for low-β context) transplanted to high-β context (Papua New Guinea, Afghanistan) should fail predictably. Failure is not random; it is a function of β-gap:

$$P(\text{failure}) = f\left(\beta_{\text{target}} - \beta_{\text{origin}}\right)$$

where $\frac{\partial P}{\partial(\beta_{\text{target}}-\beta_{\text{origin}})} > 0$.

2. **Adaptation strategies:** Successful transplants in high-β contexts should show vehicle modifications. Example: Microfinance (Grameen Bank model) succeeded in Bangladesh not by transplanting Western individual credit but by adapting to existing group-binding structures (village committees, peer pressure, collective responsibility). The content (credit access) traveled in vehicles compatible with local substrate (group loyalty, shame-based enforcement).

3. **Reverse transplants should fail symmetrically:** High-β institutions transplanted to low-β contexts should also fail. Example: Sharia courts in Western Europe generate legal pluralism tensions because they assume high-binding religious communities that low-β secular contexts reject as incompatible with individualism and state monopoly on law.

Connection to constitutional lock-in: My Constitutional Lock-In Index (CLI) should be revised:

$$CLI = f\left(\text{textual vagueness, treaty hierarchy, precedent, federal frag}, \beta, \beta_{\text{gap}}\right)$$

where β is domestic binding coefficient and β-gap measures substrate mismatch between constitutional design assumptions and implementation context.

High CLI can result from high domestic β (institutions with strong identity adhesion resist change) or high β-gap (transplanted institutions fail to replicate because substrate is incompatible). The 31.2% implementation gap in non-WEIRD contexts likely reflects both mechanisms.

Methodological implication: Legal transplant scholarship measuring "success" by formal adoption rates misses the point. Adoption without replication is memetic failure. True success requires measuring (a) behavioral compliance, (b) internalization of norms, (c) generational transmission, (d) resilience to instrumental shocks. All four depend on binding compatibility.

7.3 The paradox of persistent dysfunction

If high-binding institutions replicate better than low-binding institutions, institutional "rationality" can be evolutionarily disadvantageous. This does not mean dysfunction is desirable; it means memetic selection does not optimize for functionality.

Implication: Policy designers who ignore binding mechanisms will repeatedly experience the Rivadavia effect: rationally designed reforms that fail to propagate.

7.4 The dual challenge: Binding and embedded interests

The binding deficit operates alongside a second structural obstacle: entrenched networks of power with material interests in existing arrangements. These networks possess high binding (loyalty, tribal adhesion) *and* concrete benefits (rents, privileges, positions).

Rivadavia faced both dimensions simultaneously. His reforms lacked binding mechanisms *and* threatened the material interests of established networks. Provincial caudillos controlled territory, taxes, and militias. The Church held land, educational monopoly, and social legitimacy. Urban merchant guilds enjoyed protected markets. Each network combined psychological binding (gaucho loyalty to caudillos, faithful devotion to Church, guild solidarity) with tangible resources that rational arguments could not overcome.

Contemporary hybridizers address both dimensions:

The hybrid liberal's dual strategy: - **Binding solution:** Evangelical narratives (Purim 2024, Luna Park 2024, Decree 486/2025) provide identity vehicle for mass mobilization - **Structural attack:** "La casta" framing targets embedded networks (unions, state corporations, political class) with high binding among their members but delegitimized among the wider public

The Brazilian conservative's parallel approach: - **Binding solution:** Explicit evangelicalism, military nationalism, religious symbolism - **Structural attack:** "O sistema" rhetoric against Workers' Party networks, public sector unions, cultural establishment

The binding coefficient β thus predicts not only institutional replication capacity but also *resistance capacity* of power networks. A network with high β can defend dysfunctional arrangements against functionally superior reforms. This explains why clientelism outcompetes meritocracy in contexts where tribal loyalty (high β) dominates rational evaluation (low β).

Formal extension: Network resistance capacity R is a function of both binding β and material resources M:

$$R(\text{network}) = f(\beta, M)$$

where $\frac{\partial R}{\partial \beta} > 0$ and $\frac{\partial R}{\partial M} > 0$, and critically, β and M interact: $\frac{\partial^2 R}{\partial \beta\, \partial M} > 0$.

High binding amplifies the value of material resources (loyal followers deploy resources more effectively), and material resources amplify binding (patronage reinforces loyalty). Rivadavia faced this multiplicative effect. Contemporary liberals discovered that attacking material resources (privatization, deregulation, fiscal adjustment) *while* offering alternative binding (evangelicalism) can break the reinforcement loop.

Testable prediction: Reforms that address both dimensions simultaneously should have higher success rates than reforms addressing only binding or only structural interests.

Measuring success requires tracking (a) policy implementation persistence and (b) electoral survival under economic crisis.

7.5 Policy implications: Designing for binding

The framework generates concrete implications for institutional design and reform strategy.

For reform-minded policymakers:

Traditional rational choice institutionalism assumes that well-designed formal rules will generate compliance through incentives and sanctions. The binding deficit reveals why this fails: institutions lacking psychological infrastructure for identity adhesion collapse under the first instrumental shock.

Design principle 1: Separate content from vehicle: Reformers should not assume rational content will travel through rational vehicles. Instead, identify available high-binding vehicles in the target context (religious communities, nationalist narratives, tribal structures) and design transmission strategies accordingly.

Design principle 2: Address material resistance explicitly: Rational reforms threatening embedded networks face resistance amplified by the interaction between binding and material interests ($\frac{\partial^2 R}{\partial \beta\, \partial M} > 0$). Successful reform requires either (a) compensating displaced networks, (b) fracturing their binding (divide and conquer), or (c) offering alternative binding sources while attacking material bases.

Design principle 3: Accept hybrid incoherence: Pure programmatic coherence may be evolutionarily disadvantageous. The leader's simultaneous appeals to Judaism, evangelicalism, and Catholicism appear incoherent from a rationalist perspective but adaptive from a binding-capture perspective. The objective is not theological consistency but maximizing identity adhesion across multiple constituencies.

For researchers:

The binding coefficient β should be incorporated into institutional analysis alongside traditional variables. Constitutional Lock-In Index (CLI) calculations that ignore binding will systematically underestimate persistence of dysfunctional arrangements and overestimate adoption probability of functionally superior alternatives.

Measurement challenge: Existing datasets (World Values Survey, Latin American Public Opinion Project, Comparative Political Data Set) capture attitudes toward institutions but not binding mechanisms directly. Future surveys should include: - Identity-based versus instrumental motivations for institutional support - Resilience of support under counterfactual information about institutional performance - Generational transmission of institutional preferences independent of outcomes - Network density and composition of institutional stakeholders

For international development agencies:

Legal transplant and institutional reform programs systematically fail in non-WEIRD contexts not because beneficiaries lack capacity but because transplanted institutions lack binding substrates. A Swedish anti-corruption framework assumes low-binding civic culture (trust in abstract rules). Transplanting it to high-binding contexts (family loyalty, tribal ties) without vehicle adaptation guarantees failure.

Recommendation: International agencies should conduct binding audits before institutional transplants, mapping existing sources of identity adhesion and designing transmission strategies compatible with local psychological infrastructure. This is not cultural relativism; it is evolutionary realism.

VIII. LIMITATIONS AND FUTURE RESEARCH

8.1 Limitations

1. Anecdotal evidence: Discussed cases are illustrative, not systematic. Rigorous testing requires coded political communication analysis.

2. Endogeneity: Do leaders hybridize because they need binding, or are those with religious predisposition selected? Causality is difficult to distinguish.

3. Regional specificity: The pattern may be specific to Latin America due to its particular religious configuration. Generalization requires comparative studies.

4. Primary source verification: Several key quotes (Mises, Rivadavia) require access to primary sources for exact pagination and context confirmation. [**PRIORITY: Reality filter pending**]

8.2 Research agenda

1. Binding metric: Develop validated instrument to measure binding coefficient of political programs - Survey items capturing identity versus instrumental motivations - Experimental designs measuring adhesion resilience to negative information

2. Longitudinal study: Track support for hybrid versus pure liberal governments under economic shocks - Quarterly approval ratings crossed with economic indicators - Event studies around specific crises

3. Content analysis: Systematic coding of political communication to detect binding elements - Automated text analysis for religious references - Ritual event cataloging

4. Cross-regional comparison: Test whether pattern replicates in Eastern Europe, Asia, Africa - Are post-communist liberal parties adopting religious elements? - Do Asian liberal movements show similar dynamics?

5. Historical genealogy: Trace continuities between 19th-century Enlightenment failure and contemporary hybridizations - Detailed historiographic analysis of Generation of 1837 - Comparative study with European liberal evolution

6. Network analysis of embedded power structures: - Map concrete networks resisting reform (unions, state corporations, professional associations, territorial bosses) - Measure network properties: density, centrality, resource concentration - Test hypothesis that reform success correlates with network fragmentation - Operationalize "structural attack" effectiveness through network disruption metrics

7. Generational transmission studies: - Panel data tracking political preference inheritance across generations - Distinguish binding-based transmission (family tradition) from outcome-based adoption (instrumental evaluation) - Test prediction: high-β movements show higher parent-child preference correlation - Control for socioeconomic variables, education, geographic mobility

8. Experimental manipulation of binding elements: - Survey experiments presenting identical policy content with varying vehicles (religious framing, nationalist framing, technocratic framing, control) - Measure support levels, willingness to accept short-term costs, resilience to negative information - Test causal claim: binding vehicle increases support independent of content - Replicate across WEIRD and non-WEIRD populations to test universality

9. Computational modeling: - Agent-based models of memetic competition with binding coefficients - Simulate electoral dynamics under varying β-distributions - Test conditions under which hybridization emerges as dominant strategy - Model network resistance as function of β and material resources

10. Longitudinal case studies: - Five-year tracking of the hybrid liberal administration (2023-2028) - Measure: policy implementation rates, approval trajectories, coalition stability, evangelical community engagement - Compare predicted versus actual outcomes - Document internal tensions between rationalist and mystical components - Assess whether pastoral candidacy materializes and captures evangelical vote share

IX. CONCLUSION

9.1 The tragedy of liberal rationalism

Mises was right and wrong. Right that liberal rationalism suffered from a structural flaw: trusting that reason alone would suffice for propagation. Wrong that recognizing this flaw condemned humanity to hopelessness.

The tragedy is not that liberals failed to persuade through rational argument. The tragedy is that they defined their entire identity around the conviction that rational persuasion was the *only* legitimate method. Mises could not advocate for non-rational transmission

mechanisms without self-contradiction. His rationalist identity was not incidental; it was constitutive. To abandon it would be to abandon himself.

This explains the sixty-year silence (1949-2008) following Mises's diagnosis. Liberals understood the binding deficit but could not solve it without ceasing to be liberals as they defined themselves. The Cold War suspended this dilemma by allowing negative definition (anti-communism). The Washington Consensus suspended it by eliminating serious competition. When competition returned after 2008, liberalism faced an existential choice: evolve or die.

9.2 The evolutionary solution: Content-vehicle separation

What Mises did not anticipate, perhaps because it violated the unity of rational discourse he defended, is that memetic content and transmission vehicles are analytically separable and can be deliberately decoupled.

Liberalism can survive by riding horses not its own. The content (free markets, property rights, rule of law, fiscal discipline) can travel in vehicles borrowed from other traditions (evangelical community, religious symbolism, nationalist narrative, tribal belonging). This is not theoretical speculation; it is observable empirical reality in contemporary Latin America.

But the problem, this paper argues, is dual. Not only must rational content find non-rational transmission vehicles; it must also dismantle the entrenched networks of power that resist reform through the multiplicative interaction of binding and material interests. Caudillos, unions, state corporations, political establishments possess both tribal loyalty (β) and concrete resources (rents, patronage, coercive capacity). Rational arguments cannot overcome either dimension independently. Only simultaneous strategies addressing both can succeed.

9.3 The Argentine sequence: 200 years of recurring failure

Bernardino Rivadavia was not the first, nor will contemporary hybrid liberals be the last, to confront this dual challenge. Rivadavia's 1827 confession captured both dimensions: "talentos y voluntades" (rational design addressing the binding deficit) could not overcome "la naturaleza de las cosas" (embedded power structures with material interests).

The Generation of 1837 attempted a first correction. Sarmiento's *Facundo* (1845) is a liberal program wrapped in Romantic epic: civilization versus barbarism, the gaucho as noble savage, the pampas as stage for existential drama. Content remained rationalist (education, immigration, railroads, European institutions). Vehicle became emotional (vivid descriptions, moral struggle, narrative tension). The hybridization worked electorally, Sarmiento became president (1868-1874), but implementation faced resistance from caudillos combining gaucho loyalty with territorial control. Binding without structural attack produced partial success.

What we observe in 2023-2025 is the contemporary version of this recurring pattern, with the essential addition that both dimensions receive explicit attention: hybridization with

evangelicalism combined with aggressive attacks on established political networks. The available binding resource shifted from Romantic nationalism (now occupied by Peronism) to expanding evangelicalism (approximately 20% of population, doubled in 15 years). And competition for that resource crosses the political spectrum: libertarian right and Peronist left alike now seek evangelical constituencies.

Argentina has cycled through this problem three times in 200 years. Each time, reformers discover they need binding. Each time, they discover binding alone is insufficient. Each time, embedded networks combining loyalty and material power resist. The difference in 2023 is explicit recognition that both dimensions require simultaneous strategy.

9.4 The pattern's universality: An anthropological constant

Six cases across two centuries and three continents (French Revolution, Comte's positivism, Soviet Union, Turkish Kemalism, Argentine Enlightenment, contemporary Latin America) reveal this is not a property specific to liberalism or Latin America but an anthropological constant: rationalist projects, regardless of ideological content, face identical binding deficits.

Robespierre discovered it between the Cult of Reason (1793) and the Cult of the Supreme Being (1794). Voltaire's quote: "If God did not exist, it would be necessary to invent him." Pure reason proved insufficient; deist ritualism became necessary.

Auguste Comte discovered it when scientific positivism produced social fragmentation. His solution: Religion of Humanity (1849) with priesthood, sacraments, calendar of secular saints, prayers, rituals. Thomas Huxley called it "Catholicism minus Christianity." Comte diagnosed correctly that rational content needed emotional vehicle, even if his specific implementation failed.

The Soviet Union, the 20th century's most explicitly anti-religious ideology ("religion is the opium of the people"), developed complete political religion: Lenin's embalmed body as saint's relic, Red Corner replacing religious icons, festivals and liturgy, sacred texts, saints and martyrs. The system lasted 74 years. Binding worked even as programmatic content eventually collapsed.

Turkish Kemalism imposed radical secularization but constructed civil religion around venerated Atatürk figure. Regulation 28283 (2012) codifies wreath-laying rituals, official ceremonies, mandatory participation. A recent academic article formulates directly: "Kemalism rejects metaphysical existence on the basis of natural sciences and rationality... However, the ideology it created is almost entirely a religious organization."

This universality suggests binding deficit is not historical accident but consequence of human psychological architecture. Jonathan Haidt's moral foundations theory reached the same conclusion from experimental psychology: humans possess multiple moral foundations (care, fairness, loyalty, authority, sanctity). Programs appealing only to the first two (care, fairness) ignore most of human motivational architecture.

From EPT perspective: reason is possible component of memetic content but inefficient transmission vehicle. Memes replicating best activate multiple psychological channels. Rationalism's insistence on pure cognitive transmission is adaptive disadvantage in competitive environments.

9.5 Policy implications: Accepting evolved design

The framework generates uncomfortable implications for reformers committed to rational institutional design.

First implication: Designed coherence may be maladaptive. The leader's simultaneous appeals to Judaism (Purim March 2024), evangelicalism (Luna Park June 2024, Decree 486/2025), and Catholicism (Christmas nativity ceremony December 2024) appear incoherent from rationalist perspective. Multiple religious identities contradict theological consistency. But from binding-capture perspective, diversification is adaptive. Objective is not doctrinal purity but maximizing identity adhesion across constituencies. Coherence is constraint, not virtue.

Second implication: Rational persuasion is niche strategy, not universal solution. Enlightenment faith that reason would prevail once "superstition" was eliminated has been falsified for 230 years. Reason prevails in specific contexts (academic seminars, court proceedings, technical committees) but fails in mass politics absent high-binding vehicles. This is not failure of education or communication; it is mismatch between psychological mechanism (rational cognition) and functional requirement (mass mobilization). Attempting to "improve civic education" so citizens respond rationally to policy arguments is evolutionary equivalent of attempting to make humans photosynthesize. The substrate does not support the function.

Third implication: Institutional transplants fail predictably. WEIRD institutions assume psychological substrate (low-binding civic culture, trust in abstract rules, guilt-based morality) absent in non-WEIRD contexts. Transplanting Swedish anti-corruption framework to Papua New Guinea fails not because Papua New Guineans lack capacity but because their high-binding kinship structures generate stronger identity adhesion than abstract procedural rules. Failure rate correlates with β-gap (substrate mismatch). Development agencies should conduct binding audits before institutional transplants, mapping existing identity adhesion sources and designing transmission strategies compatible with local psychological infrastructure.

Fourth implication: Embedded networks require structural attack, not persuasion. Networks combining high binding (tribal loyalty, union solidarity, religious devotion) with material resources (rents, patronage, coercive capacity) do not yield to rational argument. Their resistance capacity R = f(β, M) where $\frac{\partial^2 R}{\partial \beta\, \partial M} > 0$ (binding and material resources interact multiplicatively). Reformers must either (a) compensate displaced networks, (b) fracture their binding (divide and conquer), or (c) offer alternative binding while attacking

material bases. Option (c), the hybrid liberal strategy, is riskiest but potentially highest return.

9.6 Normative questions beyond descriptive scope

Whether liberalism's survival through hybridization represents progress or betrayal exceeds this paper's descriptive scope. Three positions merit consideration:

Position 1 (Betrayal): Authentic liberalism requires commitment to rational persuasion as both means and end. Hybridization with religious narratives contradicts liberalism's Enlightenment foundations. A liberalism that survives by abandoning reason is not liberalism; it is something else wearing liberalism's clothes. Mises would reject this adaptation as incoherent.

Position 2 (Pragmatic evolution): Liberalism is defined by content (free markets, property rights, rule of law), not transmission vehicle. If rational vehicle proves evolutionarily maladaptive, switching vehicles preserves content. What matters is policy implementation, not aesthetic purity of transmission method. The administration implements Mises's economic program; the religious framing is instrumentally justified.

Position 3 (Synthesis): Hybridization reveals liberalism was never pure rationalism. Historical liberals (Locke, Smith, Tocqueville, Hayek) embedded their arguments in religious, moral, and cultural contexts. The fiction of pure rationalism emerged in 20th-century academic economics, not classical liberalism. The liberal-evangelical hybridization is return to liberalism's historical form, not deviation.

This paper takes no position on which view is correct. What matters empirically is that hybridization is occurring, has historical precedent, follows predictable evolutionary logic, and generates testable implications.

9.7 The research frontier

This paper establishes conceptual framework, documents historical pattern, and proposes testable predictions. Future research must move beyond case studies to systematic measurement and causal testing.

Immediate priorities:

1. **Develop validated binding coefficient instrument** (Appendix A provides draft). Pilot studies in WEIRD and non-WEIRD contexts, establish construct validity, test-retest reliability, discriminant validity from policy agreement.

2. **Code systematic content analysis** of Latin American liberal leaders' communication. Measure proportion binding versus instrumental content. Correlate with approval trajectories, electoral outcomes, policy implementation rates.

3. **Quasi-experimental comparison:** Technocratic liberal governments (secular, low binding) versus hybridized liberal governments (religious outreach, high binding) in

same national contexts. Difference-in-differences analysis of approval trajectories under economic shocks.

4. **Cross-regional replication:** Does pattern appear outside Latin America? Modi's Hindu nationalism, Duterte's Catholic populism, Erdoğan's political Islam all combine economic programs with religious binding. Are electoral and policy outcomes consistent with dual-dimension hypothesis?

Theoretical extensions:

1. **Integration with cultural evolution theory:** Henrich, Boyd, Richerson document mechanisms by which cultural practices outcompete alternatives. Binding coefficient can be formalized as component of cultural fitness functions.

2. **Connection to costly signaling theory:** Zahavy's handicap principle suggests credible signals require costs. Religious commitment involves behavioral costs (time, resources, opportunity) making it credible binding signal. Can this explain why authentic religious leaders (the evangelical pastor) entering politics have electoral viability despite policy inexperience?

3. **Network theory formalization:** Embedded power networks can be modeled as graphs where nodes (actors) have properties (β, material resources) and edges (relationships) have weights (strength of ties). Reform success becomes function of network structure, β-distribution, and resource allocation.

4. **Memetic epidemiology:** Modeling political programs as infections spreading through populations. Transmission rate depends on binding coefficient (higher β = higher R0). Existing power networks possess immunity (prior binding). Successful variants must either overcome immunity (higher β than incumbent) or attack immune hosts (structural disruption).

9.8 Final reflection: The inescapability of evolution

The central insight is not that liberalism should hybridize but that it will. Memetic selection operates with or without conscious recognition. Programs lacking binding lose fitness. Programs gaining binding gain fitness. The process is not normative; it is descriptive.

Mises recognized the fatal flaw but concluded hopelessness because solving it required abandoning his rationalist identity. He was wrong about hopelessness but right that rationalists cannot solve the problem by becoming rationalists. They solve it by ceasing to be pure rationalists.

Rivadavia could not solve the problem because Enlightenment rationalism defined his identity. Sarmiento partially solved it by incorporating Romanticism. Contemporary liberal leaders are solving it by incorporating evangelicalism while attacking embedded networks. Each adaptation reflects selection pressure; each solution is contingent on available binding sources.

The question is not whether this is desirable. The question is whether it is effective. If hybridization allows liberal economic content to replicate in conditions where pure rationalism fails, hybridization will spread. If attacks on embedded networks allow implementation where accommodation fails, attacks will spread. Moral evaluation is separate from evolutionary prediction.

Liberalism may survive the 21st century. But it will not do so as Mises envisioned: through rational persuasion of an enlightened citizenry recognizing truth through pure reason. It will do so as Rivadavia's successors discovered, as Sarmiento demonstrated, and as contemporary Latin American liberals are validating: through separation of rational content from emotional vehicles, combined with strategic dismantling of embedded interests.

Whether that represents progress, regression, or morally neutral adaptation is question for philosophers. For social scientists, what matters is that it works, it has precedent, it follows predictable logic, and its success or failure will generate evidence illuminating the evolutionary dynamics of political ideas.

The binding deficit was fatal to Mises's liberalism. It need not be fatal to liberalism. But resolution requires accepting that ideas do not compete on truth alone; they compete on replication capacity. And replication capacity, in humans, depends on psychological infrastructure rationalism explicitly rejects: identity, ritual, belonging, the irrational.

Mises was right that recognizing this reality without abandoning rationalism is impossible. He was wrong that it leaves no hope. It leaves evolutionary hope: adaptation, not through pure design but through variation, selection, and iteration. Liberalism will survive not by winning the argument but by finding better vehicles. That is not the future Mises wanted. But it may be the future liberalism gets.

REFERENCES

Aumann, R. J. (1976). Agreeing to disagree. *The Annals of Statistics*, 4(6), 1236-1239.

Comte, A. (1851-1854). *Système de politique positive, ou Traité de sociologie instituant la religion de l'Humanité*. Paris: L. Mathias.

Dawkins, R. (1982). *The Extended Phenotype: The Long Reach of the Gene*. Oxford University Press.

Dennett, D. C. (1995). *Darwin's Dangerous Idea: Evolution and the Meanings of Life*. Simon & Schuster.

Haidt, J. (2012). *The Righteous Mind: Why Good People Are Divided by Politics and Religion*. Pantheon Books.

Henrich, J. (2020). *The WEIRDest People in the World: How the West Became Psychologically Peculiar and Particularly Prosperous*. Farrar, Straus and Giroux.

Lerer, I. A. (2025). *From Transaction Costs to Memetic Fitness: Why Rational Legal Transplants Fail*. SSRN: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5750242

Lerer, I. A. (2025). *Law as Extended Phenotype: Reframing Legal Evolution Through Memetic Theory* . SSRN: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5737383

Mises, L. von. (1927). *Liberalismus*. Jena: Gustav Fischer. [English translation: *Liberalism: The Classical Tradition*, Liberty Fund, 2005]

Mises, L. von. (1949). *Human Action: A Treatise on Economics*. New Haven: Yale University Press. [Scholar's Edition, Ludwig von Mises Institute, 1998]

Sarmiento, D. F. (1845). *Facundo: Civilización y Barbarie*. Santiago de Chile: El Progreso. [English translation: *Life in the Argentine Republic in the Days of the Tyrants*, Hurd and Houghton, 1868]

Terán, O. (2008). *Historia de las ideas en Argentina: Diez lecciones iniciales, 1810-1980*. Buenos Aires: Siglo XXI Editores.

Turchin, P. (2016). *Ultrasociety: How 10,000 Years of War Made Humans the Greatest Cooperators on Earth*. Chaplin, CT: Beresta Books.

DATA SOURCES

Infobae. (2025, December 2). Evangelical pastors as political candidates in Argentina.

Letra P. (2025, December 14). Religious leaders and presidential ambitions.

Revista Noticias (Perfil). (2025, August 12). Liberal-evangelical alliances in Argentina.

PELA-USAL (Proyecto Élites Parlamentarias Latinoamericanas). Universidad de Salamanca. [**Data verification pending**]

APPENDIX A: BINDING COEFFICIENT MEASUREMENT INSTRUMENT (DRAFT)

Survey items for estimating β:

1. "I support [party/movement] because:"
   - a) Its policies will improve my economic situation (instrumental)
   - b) It represents my values and identity (binding)
   - c) Both equally
2. "If [party/movement]'s economic policies failed, I would:"

- a) Immediately withdraw support (low β)
- b) Continue supporting because it represents me (high β)
- c) Depends on severity of failure

3. "My family/community's political tradition is:"
   - a) Irrelevant to my vote (low β)
   - b) Very important to my vote (high β)
4. "I feel [party/movement] is:"
   - a) A tool to achieve goals (instrumental)
   - b) Part of who I am (binding)

Scoring: β = (binding responses) / (total responses)

Validation requirements:

The instrument must demonstrate validity across multiple dimensions before deployment in hypothesis testing.

Construct validity: β-scores should correlate with theoretically predicted behaviors. High-β respondents should exhibit: - Lower rates of party switching across elections - Higher participation in non-electoral activities (protests, donations, volunteer work) - Greater resilience to negative information about preferred party/movement - Stronger emotional reactions to party symbols, leaders, opponents

Validation approach: Behavioral tracking studies comparing survey β-scores to actual behaviors over election cycles.

Discriminant validity: β must distinguish identity adhesion from mere policy agreement. Possible confound: respondents who strongly agree with policies may score high on identity items not because of binding but because of sincere policy commitment.

Solution: Include policy agreement scale separately. Calculate partial correlation between β and behavioral outcomes controlling for policy agreement. If β predicts behavior independently, discriminant validity is established.

Test-retest reliability: β should remain stable across time periods absent major political shocks. Identity-based preferences change slowly; instrumental preferences fluctuate with news cycles.

Validation approach: Administer survey at T1, repeat at T2 (3 months later, absent major events). Calculate correlation between T1 and T2 β-scores. Target: $r > 0.70$.

Cross-cultural validity: Instrument must function in WEIRD and non-WEIRD contexts. Possible failure mode: survey response styles vary across cultures (acquiescence bias, social desirability, collectivist versus individualist framing).

Solution: Cognitive interviewing in diverse contexts. Pilot in Argentina, Brazil, United States, India, Poland. Adjust item wording to ensure equivalent construct measurement across cultures. Consider culture-specific β-sources (religious versus ethnic versus class).

Concurrent validity: β-scores should correlate with alternative binding measures: - Social network composition (proportion of friends sharing political preference) - Consumption patterns (political merchandise, media choices, social media follows) - Biographical narratives (family political traditions, formative political experiences) - Emotional intensity (self-reported affect when discussing politics)

Predictive validity: β measured at T1 should predict political behaviors at T2. High-β respondents measured before economic crisis should show lower approval decline during crisis compared to low-β respondents, controlling for economic exposure.

Sample requirements for validation: Minimum 500 respondents per country, quota sampling to ensure variation in political preferences, age, education, income. Oversampling of high-binding groups (evangelicals, Peronists, etc.) to ensure sufficient variance. Longitudinal panel structure to test temporal stability and predictive validity.

[**Note:** Full instrument validation study estimated cost: $120,000-$180,000 for five-country deployment with 18-month panel. Potential funding sources: NSF, Open Society Foundations, Inter-American Development Bank research grants.]

APPENDIX B: CASE STUDY DETAILS

B.1 French Revolution: Cult of Reason to Supreme Being (1793-1794)

Timeline: - November 10, 1793: Hébertists establish Cult of Reason; Notre-Dame cathedral transformed into Temple of Reason - November 10, 1793: Festival of Reason held at Notre-Dame with "Goddess of Reason" (actress portraying Liberty) - November 23, 1793: Paris Commune decrees closure of all churches - December 5, 1793: Robespierre denounces atheism and the Cult of Reason in speech to National Convention - March 13, 1794: Hébertist leaders arrested - March 24, 1794: Hébertists guillotined (Hébert, Momoro, Ronsin) - May 7, 1794: Robespierre's report to National Convention proposing Cult of Supreme Being - June 4, 1794: Decree establishing Cult of Supreme Being and Nature - June 8, 1794: Festival of Supreme Being at Champ de Mars (estimated 100,000+ attendees) - July 28, 1794: Robespierre guillotined (50 days after Festival)

Key quotes: - Robespierre on atheism: "Atheism is aristocratic. The idea of a Supreme Being who watches over oppressed innocence and punishes triumphant crime is altogether

popular." - Voltaire (repeated by Robespierre): "Si Dieu n'existait pas, il faudrait l'inventer" (If God did not exist, it would be necessary to invent him)

Analysis: Shift from pure rationalism (Cult of Reason, 3 weeks) to deist civil religion (Cult of Supreme Being) represents recognition that reason alone generated insufficient binding. Yet Robespierre's execution demonstrates that imposed ritual without organic community adoption fails. High-binding requirement recognized; sustainable implementation not achieved.

B.2 Auguste Comte's Religion of Humanity (1849-1857)

Structural elements: - **Object of worship:** Grand-Être (Humanity as collective entity across time) - **Priesthood:** Sociologist-priests required 14 years training (3 years mathematics/astronomy, 4 years physics/chemistry, 3 years biology, 4 years sociology/morality) - **Seven sacraments:** Presentation (naming ceremony), Initiation (age 14), Admission (age 21), Destination (career choice, age 28), Marriage, Maturity (age 63), Transformation (death), Incorporation (posthumous recognition, age 100 after death) - **Positivist Calendar:** 13 months of 28 days, each day dedicated to historical figure (Moses, Homer, Aristotle, Archimedes, Caesar, Charlemagne, Dante, Gutenberg, Shakespeare, Descartes, Frederick II, Bichat) - **Temples:** 8 circular halls representing successive sciences (mathematics, astronomy, physics, chemistry, biology, sociology, morality, synthesis) - **Prayers:** Morning prayer to Humanity, meals dedicated to family members - **Liturgical year:** Festival of Humanity, Festival of Woman, Festival of Priests

Membership data: - 1857: Positivist Society of Paris (estimated 40-50 members at peak) - 1870s-1880s: Positivist churches established in London (1867), Rio de Janeiro (1881), Chile (1877) - Brazil: Most successful adoption; influence on Republican movement, national motto "Ordem e Progresso" derived from Comte - 1900: Estimated total global membership 5,000-10,000 (primarily Brazil, France, England)

Why it failed: Ritual complexity without organic community. Comte designed every detail consciously (architecture, clothing, prayers), but religion emerges from practice, not blueprint. Lacked embedded social networks that could sustain transmission across generations. High-binding elements present but artificially constructed rather than organically adopted.

B.3 Soviet Political Religion (1917-1991)

Quantitative indicators: - **League of Militant Atheists:** 5 million members by 1932 (peak), 300,000 by 1940 (decline after Stalin turned toward Orthodox Church during WWII) - **Lenin Mausoleum visits:** Estimated 10 million visitors 1924-1940, 2.5 million annual visitors during Soviet era peak (1960s-1970s), currently 2 million per year post-Soviet - **Churches closed:** 1917 baseline ~54,000 Orthodox churches; by 1939 ~500 remained open (99% closure rate); by 1988 ~7,000 reopened - **Red Corner prevalence:** 1930 census data suggests 60-70% of urban households maintained Red Corner with Lenin/Stalin images; rural adoption lower (~30-40%)

Ritual institutionalization: - **Mandatory festivals:** May 1 (International Workers' Day), November 7 (October Revolution), March 8 (International Women's Day), Lenin's Birthday (April 22) - **Parades:** Moscow Red Square parade involved 25,000-40,000 participants, 500,000-1,000,000 spectators - **Komsomol initiation:** Youth organization enrollment ceremony at age 14, sworn oath to Communist Party, red neckerchief symbol (1920s-1980s enrollment rates 70-80% of eligible youth) - **Pioneer organization:** Children's organization ages 10-15, ritual initiation ceremony, oath-taking, symbol (red neckerchief), salute, camps, activities (1922-1991 enrollment peak 25 million in 1970s)

Theoretical significance: Soviet case demonstrates that high binding can sustain political system despite economic dysfunction (chronic shortages, technological lag, productivity stagnation). Binding coefficient β maintained institutional fitness F even as functional performance P declined. System collapsed when Gorbachev's reforms (glasnost, perestroika) inadvertently weakened binding mechanisms (questioning of ideological foundations) without replacing them, reducing both β and M (material coercion) simultaneously.

B.4 Turkish Kemalism (1923-present)

Ritual codification: - **Regulation 28283 (2012):** Specifies wreath-laying ceremonies at Atatürk monuments, official attendance requirements, wreath specifications (size, materials, ribbon text), presiding authorities, ceremonial order - **Mandatory education:** Atatürk's Principles taught in primary/secondary schools; university courses required until 1990s - **Minute of silence:** Daily 9:05am moment of silence on November 10 (anniversary of Atatürk's death at 9:05am 1938); radio/TV broadcasts pause, traffic stops, sirens sound - **Monuments:** 3,000+ Atatürk statues nationwide; defacing statue is criminal offense (Law No. 5816, 1951)

Electoral dynamics: - **1923-1950:** Single-party rule (Republican People's Party, Kemalist) - **1950-2002:** Multi-party era but Kemalist military interventions (1960, 1971, 1980, 1997 "postmodern coup") when elected governments deviated from secularism - **2002-present:** AKP (Justice and Development Party, Islamic-rooted) electoral dominance; gradual erosion of Kemalist institutions - **Erdoğan strategy:** Did not frontally attack Kemalism; instead offered alternative binding (political Islam, Ottoman nostalgia, neo-Ottomanism) while gradually weakening Kemalist power bases (military purges after 2016 coup attempt, judicial reforms, media control)

Theoretical significance: Kemalism demonstrates that secular civil religion can generate binding independent of supernatural belief. Cult of personality (Atatürk as national savior), ritualized practices (wreath-laying, moment of silence), sacred symbols (flag, anthem, founding narrative), and boundary maintenance ("enemies of Atatürk" as ultimate political insult) functioned as binding mechanisms for 80+ years. Erdoğan's success required offering competing binding (Islamic identity), not defeating Kemalism through rational argument.

B.5 Argentine Evangelical Growth 2008-2024

Demographic trajectory: - **1960:** <2% evangelical (estimate based on church records) - **1980:** ~4% evangelical (ARDA estimates) - **2008:** ~9% evangelical (first systematic national survey, CONICET) - **2019:** ~15.3% evangelical (national census religious affiliation question) - **2024:** ~20% evangelical (Aciera projections based on church membership growth rates)

Geographic concentration: - **Highest:** Greater Buenos Aires periphery (20-25%), Northwest provinces (Salta, Jujuy, 18-22%), Northeast (Chaco, Formosa, 22-25%) - **Lowest:** CABA (Buenos Aires city center, 12-15%), Patagonia (10-12%) - **Urban versus rural:** Urban areas show faster growth (2008-2024 doubling) versus rural stability

Denominational breakdown: - **Pentecostal:** ~65% of evangelicals (Assemblies of God largest single denomination, ~2.5 million members) - **Baptist:** ~15% (~1.3 million members) - **Independent/neo-Pentecostal:** ~12% (mega-churches like the evangelical pastor's Rivendale, Presencia de Dios, etc.) - **Other Protestant:** ~8% (Methodist, Presbyterian, Lutheran, etc.)

Political institutionalization timeline: - **2019:** Evangelical legislators informally coordinate but no formal bloc - **2021:** Evangelical Interparliamentary Group formed (cross-party, 15 members Chamber + 3 Senators) - **2023:** First explicitly confessional evangelical legislative bloc (8 deputies La Libertad Avanza, 3 PRO, 2 Juntos por el Cambio provincial parties) - **2024:** Evangelical caucus grows to 22 deputies (5% of Chamber) after mid-term provincial elections

Argentine liberal-evangelical engagement timeline: - **March 13, 2024:** Purim celebration with Orthodox Jewish community, Jabad Lubavitch Buenos Aires (estimated 300-400 attendees) - **June 4, 2024:** Luna Park evangelical mega-event organized by Aciera (Alianza Cristiana de Iglesias Evangélicas de la República Argentina), the hybrid liberal keynote speech (estimated attendance 15,000-18,000) - **July 22, 2025:** Decree 486/2025 published, grants legal personhood to non-Catholic religious organizations (evangelical churches primary beneficiaries) - **August 2024:** Meeting with Evangelical Missionary Union, commitment to educational reforms favoring religious schools - **December 24, 2024:** Presidential Grenadier Guards transfer Baby Jesus image to official nativity scene, Casa Rosada ceremony

CB Consultora poll (December 2025) 2027 presidential scenarios: - **the hybrid liberal:** 32.1% - **Cristina Fernández de Kirchner:** 18.7% - **Axel Kicillof:** 14.3% - **Mauricio Macri:** 11.2% - **Horacio Rodríguez Larreta:** 8.9% - **the evangelical pastor:** 1.8% - **Others/Undecided:** 13.0%

Analysis: the evangelical pastor's 1.8% represents instant name recognition and voting intention for first-time candidate with zero political experience, based purely on religious leadership. Comparable entry-level polling for the hybrid liberal (2021 first poll inclusion) was 2.3%. Suggests religious binding translates to immediate electoral viability independent of programmatic content.

B.6 Comparative Table: Binding Coefficient Estimates (β-scores)

Case	Period	Estimated β	Evidence basis
Cult of Reason	1793 (3 weeks)	$\beta \approx 0.15$	Low adhesion, rapid collapse, forced participation
Cult of Supreme Being	1794 (50 days)	$\beta \approx 0.35$	Imposed ritual, attendance but minimal internalization
Comte Religion of Humanity	1849-1900	$\beta \approx 0.40$	Dedicated core followers, no mass adoption
Soviet political religion	1925-1953	$\beta \approx 0.65$	Generational transmission, ritual compliance, identity markers
Soviet post-Stalin	1953-1985	$\beta \approx 0.50$	Declining but persistent; ideology maintained despite cynicism
Turkish Kemalism	1923-1980	$\beta \approx 0.70$	Military enforcement, educational saturation, ritual ubiquity
Turkish Kemalism	1980-2010	$\beta \approx 0.55$	Erosion through globalization, Islamic revival
Argentine Peronism	1945-present	$\beta \approx 0.75$	Strongest case: multi-generational identity, survives policy failures
Macri PRO	2015-2019	$\beta \approx 0.25$	Management party, no identity vehicle, collapsed under recession
the hybrid liberal La Libertad Avanza	2023-present	$\beta \approx 0.55$ (rising)	Hybrid binding under construction, evangelical + charismatic elements

Methodological note: β-scores are illustrative estimates, not precise measurements, based on available qualitative and quantitative indicators (ritual compliance rates, generational transmission, resilience to instrumental failures, identity salience in self-descriptions). Future research requires validated survey instruments (Appendix A) to generate rigorous quantitative β-scores.

Disclaimer: Mathematical formalizations were developed with AI assistance to express conceptual arguments in formal notation. The author assumes full responsibility for theoretical claims and empirical interpretations.